Intro to Data Science and Big Data Analytics
Lab 1
Descriptive Analytics for Business

1 Key Metrics Analysis

Key Metrics or Key Performance Indicators (KPIs) are the main method that we use to measure the performance of a company, division, department or team. Key Metrics provide an overall understanding of how the business is performing. Key Metrics usually include the following types of questions:

a) What is the Total Sales?

b) What is the Total Profit?

c) What is the Profit Ratio?

d) How many transactions were there processed?

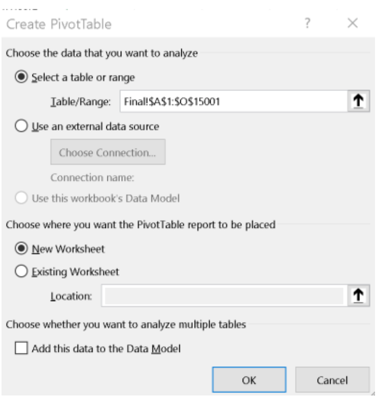
e) What was the average Sales amount?

f) What was the highest Sales amount?

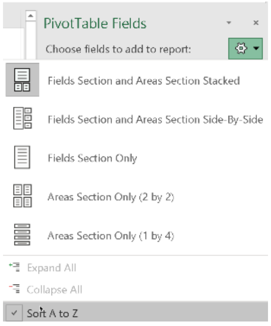
g) What was the lowest Sales amount?

File to use: `SalesData.xlsx`

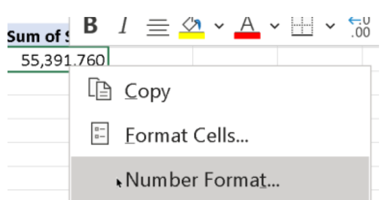
Create a Pivot table from the current range of data.
Make sure you have selected a cell in the range.
Insert -> Pivot table
The Create Pivot table should automatically detect the complete range of cells with valid values.
Make sure you create the Pivot table in a new worksheet.

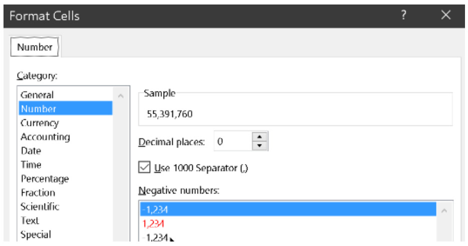


You can change the order of the Pivot Table fields as they appear in the main list: its sometimes useful to sort them in alphabetical order if you have many fields.

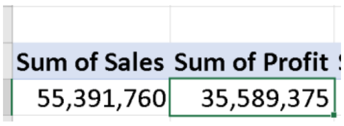


Select the Sales field to get the Sum of Sales, then perform a Number Format to make it easier to view.

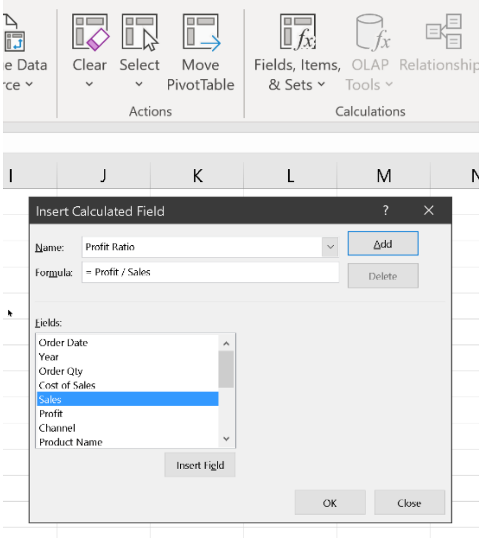


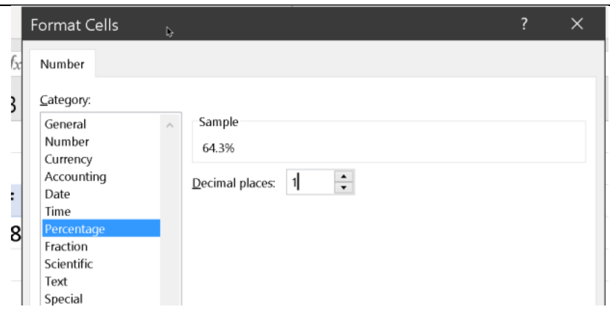


Repeat this with the Profit field to get the Sum of Profit, with a similar Number Format as well.



We can now create a Calculated Field to obtain the Profit Ratio.
With at least once cell selected in the pivot table, select Pivot Table Analyze from the top main menu, select Fields, Items and Sets
Next insert a Calculated Field to obtain the Profit Ratio, and format the result accordingly

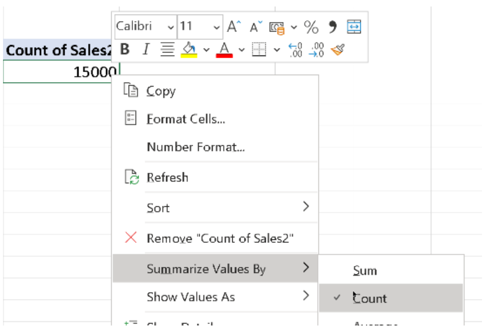




Sum of Sales	Sum of Profit	Sum of Profit Ratio
55,391,760	35,589,375	64.3%

To count the number of transactions, we would typically use a unique Sales ID, which we don't have here in this dataset. We can initially use the Sales Field to start with (drag and drop into the Values area to obtain a new Sum of Sales 2).

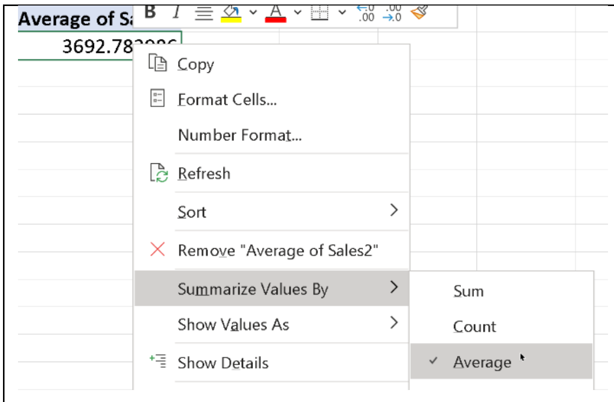
Then select this cell and from the context menu, change it to Summarize Values by Count to get the actual number of transactions (which we can also double confirm from the source dataset)



Sum of Sales	Sum of Profit	Sum of Profit Ratio	Count of Sales2
55,391,760	35,589,375	64.3%	15000

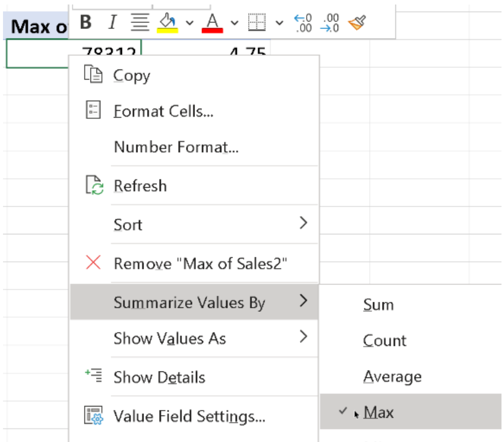
To get the average Sales, use the Sales field again in the same manner (drag and drop into the Values area to obtain a new Sum of Sales 2).

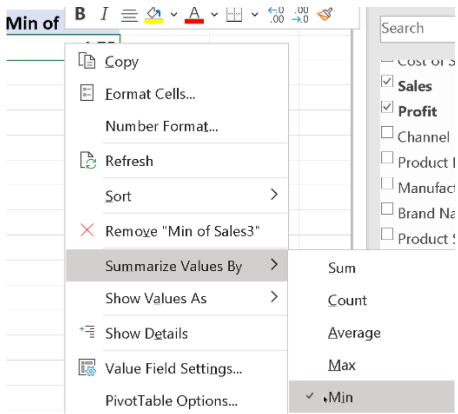
Then select this cell and from the context menu, change it to Summarize Values by Average to get the average Sales value, then format it accordingly.



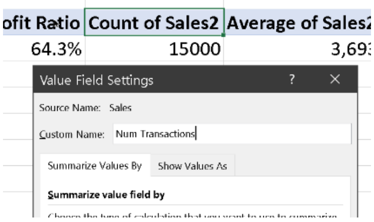
Count of Sales2	Average of Sales2
15000	3,693

Add two more Sales Fields into the Values area in a similar manner as previously, and then perform a Summarize Values by Min and Max for these two new fields and format them appropriately.





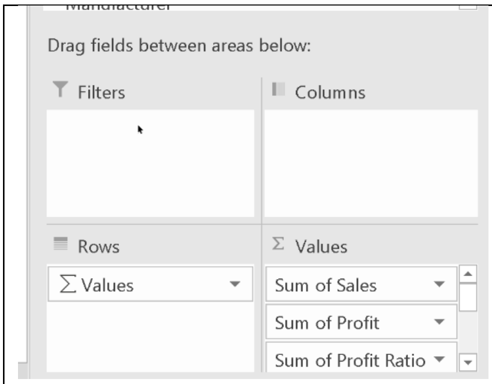
Now that we have all our numbers, you can double click on any (or all) of the field headers to change the field names to something more appropriate if you wish.

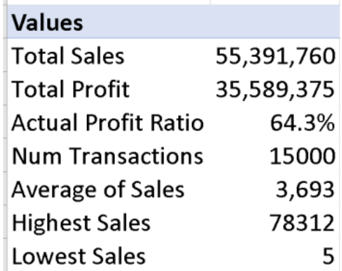


We now have our key metrics listed out in a row:

Sum of Sales	Sum of Profit	Sum of Profit Ratio	Num Transactions	Average of Sales	Highest Sales	Lowest Sales
55,391,760	35,589,375	64.3%	15000	3,693	78312	5

You can transpose the rows and columns for a better layout:

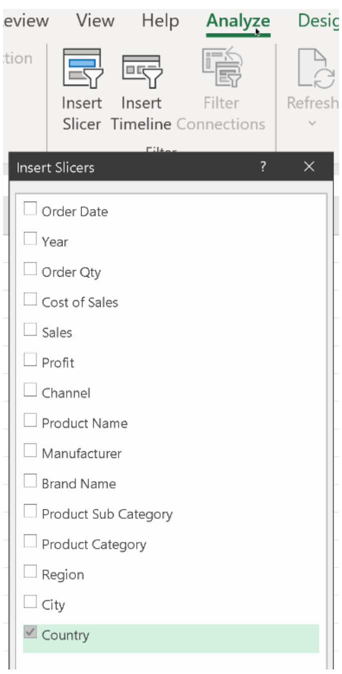


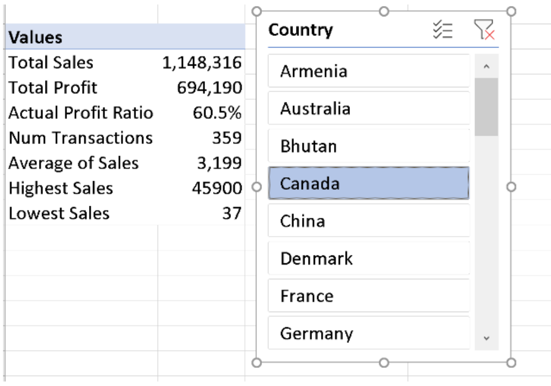


Values	
Total Sales	55,391,760
Total Profit	35,589,375
Actual Profit Ratio	64.3%
Num Transactions	15000
Average of Sales	3,693
Highest Sales	78312
Lowest Sales	5

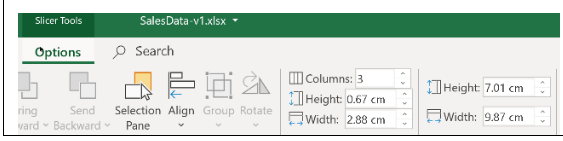
To zoom in and get more details on the metrics based on a certain category (for e.g. country), we can filter the table to decide which subset of records to apply the various aggregation operations we have performed so far on.

For this purpose, we can use a slicer. Go to Pivot Table Analyze -> Insert Slicer, and select Country. You can then select particular countries from the slicer drop down to get the metric values only for sales records related to that particular country.

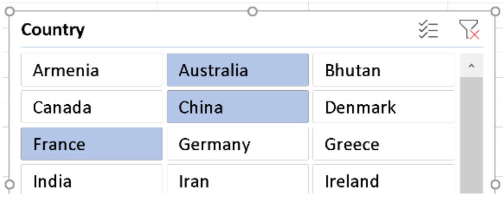


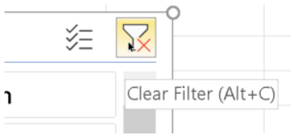


With the slicer selected and then selecting the Slicer option in the main menu, you can fine tune the slicer by adjusting the number of columns, as well as their height and width (you can just click and drag on the slicer itself to manually change this) to make them easier to work with:

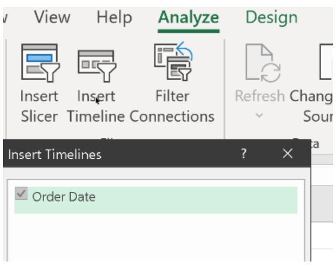


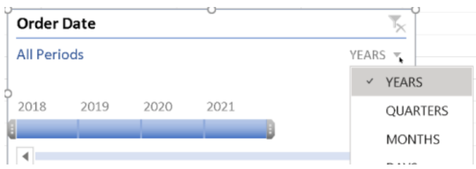
You can use the Ctrl+Left mouse click to select multiple countries at the same time, so the aggregation operations will apply to all these countries. You can then select the clear Filter button to clear the filter and make the aggregation apply to the entire dataset.



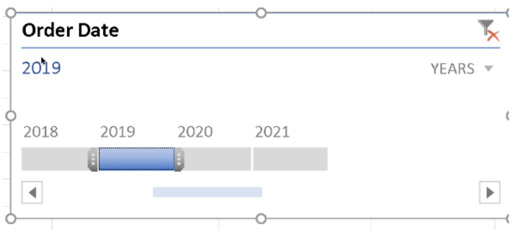


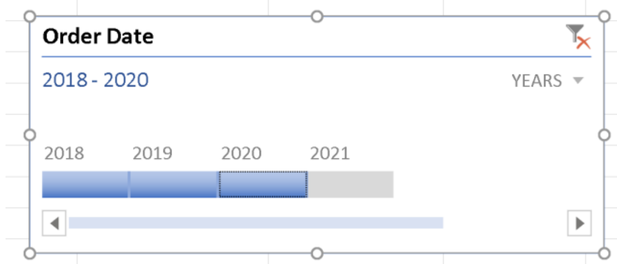
You can now also insert a Timeline and decide which particular period (Years, Quarters, Months) to examine.



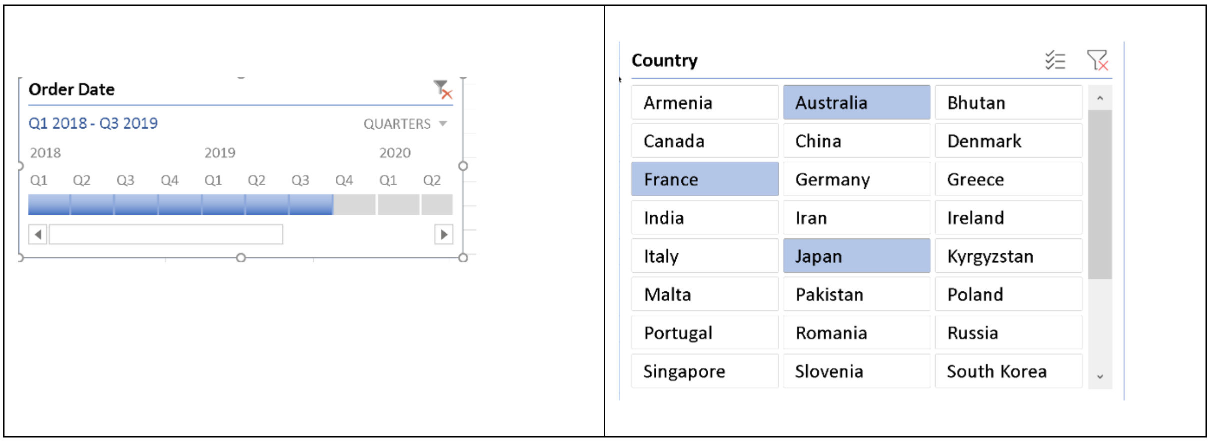


You can select a particular period or a range of periods (Shift-Click):

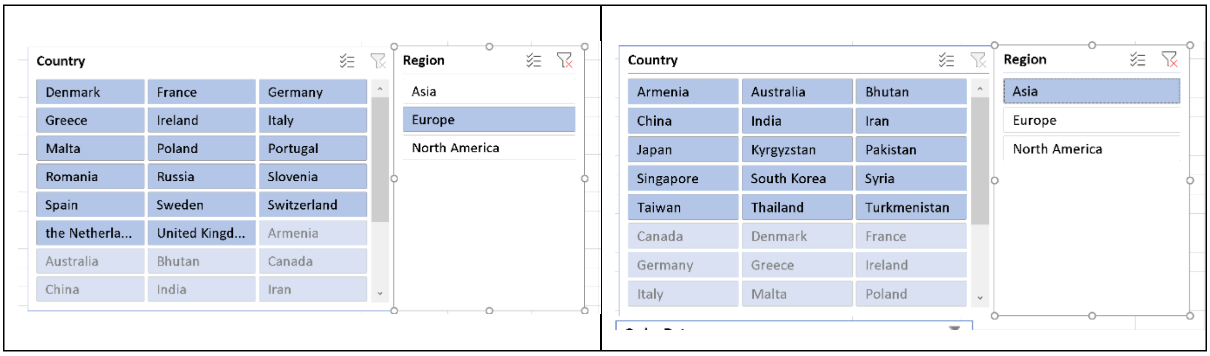




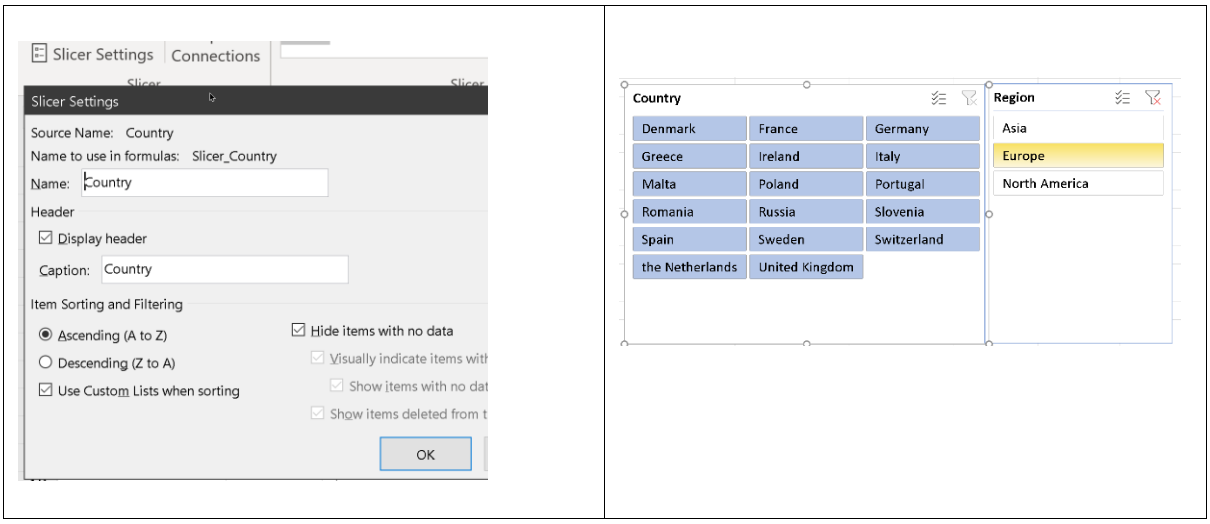
You can also select items from both filters to combine their filtering action together, for e.g. to perform aggregation operations on a specific country (or group of countries) over a specific period of time.



You can introduce another slicer for Region, and notice that selection of items in either slicer for Country or Region will influence the other: based on which countries are included in which region. To see this properly in effect, you will have to clear the filter in either one or both of the slicers.



You can also set the Slicer Settings to only clearly show the countries within a particular region, and not just grey out the countries that are not in that region (to make the analysis even more clearer):



2 Practical Exercise for Key Metrics Analysis

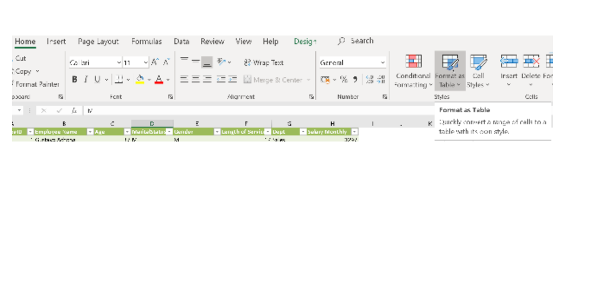
File to use: `EmployeeData.xlsx`

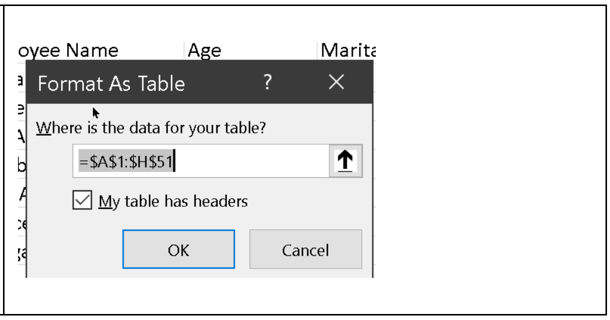
Analytical activities to perform:

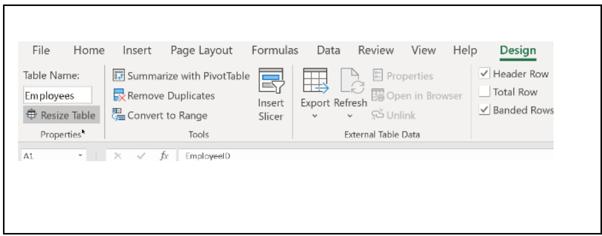
1. How many employees are in John's company?
2. What is the average age of the employees?
3. What is the highest age at the company?
4. What is the lowest age at the company?
5. What is the average Length of Service?
6. What is the longest Length of Service?

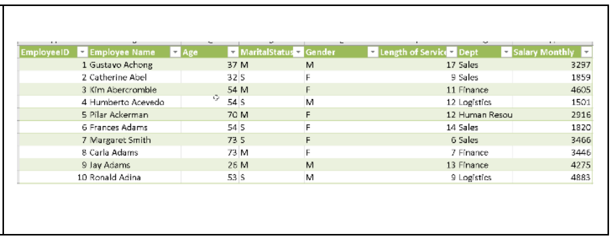
Drill down further into this information based on the following categories: Dept and Gender

You can initially format the range of cells of the original data set as a table to make it easier to work with, and to create a Pivot Table. To do this, select a cell in the data range, go to Home -> Format as Table, format it and give it an appropriate name.



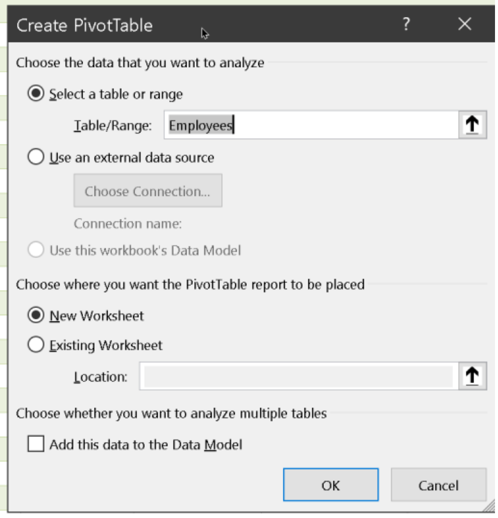






EmployeeID	Employee Name	Age	MaritalStatus	Gender	Length of Service	Dept	Salary Monthly
1	Gustavo Achong	37	M	M	17	Sales	3297
2	Catherine Abel	32	S	F	9	Sales	1859
3	Kim Abercrombie	54	M	F	11	Finance	4605
4	Humberto Acevedo	54	S	M	12	Logistics	1501
5	Pilar Ackerman	70	M	F	12	Human Resou	2916
6	Frances Adams	54	S	F	14	Sales	1820
7	Margaret Smith	73	S	F	6	Sales	3466
8	Carla Adams	73	M	F	7	Finance	3446
9	Jay Adams	26	M	M	13	Finance	4275
10	Ronald Adina	53	S	M	9	Logistics	4883

Once done, you can generate a Pivot Table in the usual way, but this time referencing the Table.


Create PivotTable
?
Choose the data that you want to analyze
Select a table or range
Table/Range: Employees
Use an external data source
Choose Connection...
Connection name:
Use this workbook's Data Model
Choose where you want the PivotTable report to be placed
New Worksheet
Existing Worksheet
Location:
Choose whether you want to analyze multiple tables
Add this data to the Data Model
OK
Cancel